# Well-quasi-ordered classes of bounded clique-width main 63025d0c2bb13d7ba1f53e4ace38ba72f7888dc2 2026-01-08 10:38:40 +0100

Aliaume Lopez
aliaume.lopez@bordeaux-inp.fr
INP Bordeaux, LaBRI, CNRS
Bordeaux, France

Maël Dumas
mdumas@mimuw.edu.pl
University of Warsaw
Warsaw, Poland

## ABSTRACT

We study classes of graphs with bounded clique-width that are well-quasi-ordered by the induced subgraph relation, in the presence of labels on the vertices. We prove that, given a finite presentation of a class of graphs, one can decide whether the class is labelled-well-quasi-ordered. This solves an open problem raised by Daligault, Rao and Thomassé in 2010. From our proof techniques, we also derive (restricted versions of) conjectures of Pouzet regarding well-quasi-ordering of graphs under the induced subgraph relation. Finally, we provide a structural characterization of those classes as those that are of bounded clique-width and do not existentially transduce the class of all finite paths.

## CCS CONCEPTS

• **Formal languages and automata theory**; • **Graph theory**;

## KEYWORDS

well-quasi-ordering, clique-width, automata theory, monoids, factorization forests, gap embedding

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This document uses knowledge: a notion points to its *definition*.

## 1 INTRODUCTION

The theory of well-quasi-orderings (WQOs) provides a powerful combinatorial setting that has found applications in various areas of mathematics and computer science. In graph theory, the celebrated result of Robertson and Seymour [20] states that the class of all finite graphs is well-quasi-ordered under the minor relation, a profound result with deep algorithmic consequences. WQOs are also at the heart of *well-structured transition systems*, infinite state transitions systems over which verification algorithms can be designed [2, 3]. One of the appeal of WQOs is that they are closed under various operations, such as the sum and the product of WQOs. As an example, the closure of WQOs under the finite words, also known as Higman's lemma [14], has been used in the verification of so-called *lossy channel systems* [1].

Undirected finite graphs are naturally equipped with the *induced subgraph relation*, where $G$ is an induced subgraph of $H$ if $G$ can be obtained from $H$ by deleting vertices. Unlike the graph minor relation, the induced subgraph relation is not a well-quasi-ordering on the class of all finite graphs, as witnessed by the infinite family of cycles of increasing size, which form an infinite sequence of pairwise incomparable graphs. However, some classes of finite graphs are well-quasi-ordered under the induced subgraph relation: for instance, the class of all finite paths, or the class of all finite cliques. Distinguishing which classes of (finite) graphs are well-quasi-ordered under the induced subgraph relation is a long-standing open problem in graph theory, dating back to Pouzet's seminal work [18].

In general, there is little hope to characterise all such classes, since any countable order with finite infixes can be embedded into the induced subgraph relation on finite graphs (this is a consequence of [16, Lemma 5.1]).

Mael: It does not seems that usual? Next sentence sounds weird. Maybe we should say something like "we can enrich the definition to say something" ?

Fortunately, one usually considers *labelled* well-quasi-orderings on classes of finite graphs. Given a class $C$ of finite graphs, and a well-quasi-ordering $(X, \leq)$, one can consider the class $\mathsf{Label}_X(C)$ of graphs in $C$ whose vertices are labelled by elements of $X$, that is, equipped with a function $\ell\colon V(G) \to X$. The induced subgraph relation is then extended to labelled graphs by requiring that the labels are preserved by the embedding, i.e. that the label of a vertex in the smaller graph is less than or equal to the label of its image in the larger graph. A class $C$ of finite graphs is said to be *$(X, \leq)$-well-quasi-ordered* if the class $\mathsf{Label}_X(C)$ is well-quasi-ordered under the labelled induced subgraph relation. We will say that it is *$k$-well-quasi-ordered* if it is $(X, \leq)$-well-quasi-ordered where $X$ is a $k$-element antichain, and that it is *$\forall$-well-quasi-ordered* if it is $(X, \leq)$-well-quasi-ordered for every well-quasi-ordering $(X, \leq)$.

*Example 1.1.* The class Cycles of all finite cycles is not well-quasi-ordered, since the infinite sequence of cycles forms an infinite antichain. The class Paths of all finite paths is well-quasi-ordered since $(\mathsf{Paths}, \subseteq_i)$ is isomorphic to $(\mathbb{N}, \leq)$. However, it is not 2-well-quasi-ordered, since the infinite sequence of paths with colored

Authors' addresses: Aliaume Lopez, aliaume.lopez@bordeaux-inp.fr, INP Bordeaux, LaBRI, CNRS, Bordeaux, France; Maël Dumas, mdumas@mimuw.edu.pl, University of Warsaw, Warsaw, Poland.

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| Class | WQO | 2-WQO | ∀-WQO |
|---|---|---|---|
| Cycles | × | × | × |
| Paths | ✓ | × | × |
| Cliques | ✓ | ✓ | ✓ |

**Figure 1: Well-quasi-ordering properties of classes in Example 1.1.**

endpoints forms an infinite antichain. The class Cliques of all finite cliques is ∀-well-quasi-ordered, since any $X$-labelled clique can be identified with a finite multiset of labels from $X$, and the set of finite multisets over a well-quasi-order is itself well-quasi-ordered by standard results on well-quasi-orders [10].

The study of labelled structures is a recurring theme in the theory of well-quasi-orderings: classical results such as Higman's lemma [14] or Kruskal's tree theorem [15] are actually stating that some classes of finite relational structures (respectively finite linear orders and finite meet trees) are well-quasi-ordered under the labelled embedding relation, for every well-quasi-ordering of labels. It was conjectured by Pouzet [18] that being ∀-well-quasi-ordered reduces to being 2-well-quasi-ordered for hereditary classes of structures (see [19, Problems 9, (1)] for a modern formulation). Another interesting conjecture of Pouzet [19, Problems 12] states that when a class $C$ is ∀-well-quasi-ordered, one can add a total order on the vertices of its structures such that the new class is still ∀-well-quasi-ordered. This conjecture gained interest in the recent development of algebraic methods for polynomials over relational structures [13, Section 3, Theorem 11 and 12], where "monomials" are defined as structures labelled by $(\mathbb{N}, \leq)$. We refer to this property as being *orderably ∀-well-quasi-ordered*.

Even for classes of finite undirected graphs, these conjectures remain open. Several works dating back to the 90s have idendified structural properties of hereditary classes of finite graphs that guarantee that they are ∀-well-quasi-ordered, such as being of bounded tree-depth [12]. In 2010, Daligault, Rao and Thomassé [9] initiated the study in the reverse direction, by considering a syntax for graph classes (using NLC graph expressions), and identifying which ones among them are 2-well-quasi-ordered. In this context, they verified Pouzet's conjectures, but were unable to generalise their results to arbitrary arbitrary graph classes [9, Conjecture 4]. All the classes considered in [9] have so-called *bounded clique-width*,[1] and it is believed that it is enough to understand hereditary classes of bounded clique-width to solve these conjectures in full generality, as stated in the following conjecture.

arbitrary sub-classes ?

Conjecture 1.2 ([9, Conjecture 5]). *Let $C$ be a hereditary class of finite graphs of that is 2-well-quasi-ordered. Then, $C$ is of bounded clique-width.*

On parallel with the conjectures of Pouzet, another line of research has focused on finding structural obstructions to being ∀-well-quasi-ordered. The natural candidate for such an obstruction is the class of all finite paths Figure 1. For the particular classes studied in [9], it was shown that every class that is not 2-well-quasi-ordered contains all finite paths [9, Corollary 2]. There are examples of hereditary classes that are not 2-well-quasi-ordered and do not contain all finite paths, but on every known such example, one can "extract" finite paths using very simple logical formulas. This lead to the following conjecture, where existentially transduces is a notion from model theory (see Section 2 for a formal definition).

Conjecture 1.3 ([17, Conjecture 26]). *Let $C$ be a hereditary class of finite graphs. If $C$ is not 2-well-quasi-ordered, then it existentially transduces the class of all finite paths.*

Another kind of obstruction to being ∀-well-quasi-ordered comes from the existence of infite antichains with a periodic structure that were called *periodic antichains* in [4, Section 7].

Conjecture 1.4 ([4, Conjecture 2]). *Let $C$ be a hereditary class of finite graphs that is not 2-well-quasi-ordered. Then, there exists an infinite (weak) periodic antichain in $C$.*

Mael: Is it really new or does it build on existing tools ?

In 2024, Lopez [17] introduced a new technique to study well-quasi-orderings on classes of graphs of bounded *linear* clique-width, that leveraged combinatorial tools from automata theory (Simon's factorisation forests [21]). This allowed to prove a weakening of Pouzet's conjectures [17, Corollary 2], namely, that a hereditary class of bounded linear clique-width is ∀-well-quasi-ordered if and only if it is $k$-well-quasi-ordered for every $k \in \mathbb{N}$. It is apparent from the proof that the techniques also yield a natural ordering on the vertices whenever the class is ∀-well-quasi-ordered, although this is not formally stated in the paper. It was conjectured [17, Section 5] that these techniques could be refined to work on classes of bounded clique-width (not necessarily linear), and actually obtain the correspondence between 2-well-quasi-ordering and ∀-well-quasi-ordering.

*Contributions.* Our main contribution is Theorem 1.5, that answers positively to the conjectures of [17]. To that end, we first consider generalisation of the framework of [9] from NLC graph expressions to MSO-interpretations from trees (these will be formally defined in Section 2). This also generalises the setting of [17] that studied MSO-interpretations from *words* (i.e., paths). Our main combinatorial result is the following.

Theorem 1.5. *Let $C$ be the image of a class of finite trees under an MSO interpretation $\mathcal{I}$. Then, the following are equivalent and decidable given $\mathcal{I}$:*

(1) *$C$ is 2-well-quasi-ordered,*
(2) *$C$ is ∀-well-quasi-ordered,*
(3) *$C$ is orderably ∀-well-quasi-ordered,*

From Theorem 1.5, we derive the following corollary that solves Pouzet's conjectures for hereditary classes of bounded clique-width.

Corollary 1.6. *The same holds when $C$ is a hereditary class of bounded clique-width graphs.*

Because our proof techniques exhibit structural obstructions to being 2-well-quasi-ordered, we also obtain the following characterisations in Theorem 1.7. These answer positively to the conjectures of [4, 17]. In particular, for a hereditary class $C$ to be 2-well-quasi-ordered, it suffices that it avoids very simple (strong) periodic antichains. and weak ?

[1] This will be defined formally in Section 2

Theorem 1.7. *Let $\mathcal{C}$ be a hereditary class of finite graphs of bounded clique-width. Then, the following are equivalent:*

(1) *$\mathcal{C}$ is not 2-well-quasi-ordered,*
(2) *There exists a class $\mathcal{D} \subseteq \mathcal{C}$ of bounded linear clique-width which is not 2-well-quasi-ordered,*
(3) *There exists a strong periodic antichain in $\mathcal{C}$,*
(4) *There exists a weak periodic antichain in $\mathcal{C}$,*
(5) *$\mathcal{C}$ existentially transduces the class of all finite paths.*

The results of Corollary 1.6 extend to non-hereditary classes of bounded clique-width graphs, but for the sake of clarity and space, we only state them on hereditary ones.

Mael: C'est peut-être un peu trop fort de dire ça, tu prévois d'essayer de le justifier quelque part ?

On the other hand, Theorem 1.7 does not immediately extend to non-hereditary classes, as witnessed by the following example.

*Example 1.8.* The class $\mathcal{C}$ of complete binary trees has bounded clique-width, is not 2-well-quasi-ordered, but contains no periodic antichain, and every subclass $\mathcal{D} \subseteq \mathcal{C}$ of bounded linear clique-width is 2-well-quasi-ordered. It happens that $\mathcal{C}$ existentially transduces the class of all finite paths, but it does not existentially interprets them.

Proof. It is well-known that the class of complete binary trees has bounded clique-width. Furthermore, it is not 2-well-quasi-ordered, since one can color the leaves and the root of the complete binary trees to form an infinite antichain. On the other hand, any subclass $\mathcal{D} \subseteq \mathcal{C}$ of bounded linear clique-width must be finite: indeed, complete binary trees have unbounded linear clique-width. Finally, because periodic antichains have bounded linear clique-width (see Section 6) there cannot be any periodic antichain in $\mathcal{C}$.

The existential transduction of all finite paths from $\mathcal{C}$ is obtained by transducing all induced subgraphs, that in particular contains all finite paths. There cannot be an existential interpretation of all finite paths from $\mathcal{C}$, because paths have only two automorphisms (reversing and identity), while complete binary trees have many more, and interpretations must preserve automorphisms. □

ref ?

*Outline.* Section 2 introduces the necessary background on well-quasi-orders, graphs, MSO interpretations and first-order transductions. Section 3 shows how to reduce the study of graph classes obtained by MSO interpretations from trees to the study of trees labelled over finite monoids. This section also shows how to leverage forward Ramseyan splits to obtain well-behaved tree representations of graphs, that are well-quasi-ordered (as trees) with respect to the so-called *gap embedding* relation. Section 4 identifies combinatorial obstructions to being 2-well-quasi-ordered in images of interpretations from trees, and is the core combinatorial part of the paper. Section 5 brings together the previous results to prove our Theorem 1.5 and Corollary 1.6. We also discuss the equivalence between the first two items of Theorem 1.7. The remaining items of Theorem 1.7 are proven equivalent in Section 6. Finally, in Section 7 we discuss some perspectives and open problems.

## 2 PRELIMINARIES

We assume the reader to be familiar with basic notions of logic on finite relational structures, such as monadic second-order logic (MSO), first-order logic and quantifier-free formulas.

Mael: TODO: unify $(u, v)$ and $uv$)

Mael: Remarque globale : si tu dis qu'on ne parle que de graphe/arbre fini, alors pas besoin de repréciser fini à chaque fois.

*Graphs.* A *graph* is a pair $G = (V, E)$ where $V$ is a set of *vertices* and $E \subseteq V \times V$ is a set of *edges*. A graph is *undirected* if $(u, v) \in E$ implies $(v, u) \in E$ for all $u, v \in V$ . In this paper we will focus on finite undirected graphs. Given a set $X$, an *$X$-labelled graph* is a graph equipped with a function $\ell \colon V \to X$ that assigns a label to each vertex. Given a class $\mathcal{C}$ of finite undirected graphs, write $\mathrm{Label}_X(\mathcal{C})$ for the class of freely $X$-labelled graphs in $\mathcal{C}$, i.e., the class of all $X$-labelled graphs $G = (V, E, \ell)$ such that $G \in \mathcal{C}$.

, directed else ??

without self-loop ?

Whenever $(X, \leq)$ is a quasi-order, we say that a graph $G$ *embeds* into a graph $H$ if there exists a *monomorphism* from $G$ to $H$, i.e., an injective function $f \colon V_G \to V_H$ such that for all $u, v \in V_G$, $(u, v) \in E_G$ if and only if $(f(u), f(v)) \in E_H$, and such that for all $v \in V_G$, we have $\ell_G(v) \leq \ell_H(f(v))$. We write $G \subseteq_i H$ to denote that $G$ embeds into $H$, which we also call the *induced subgraph relation*. When no order is specified along a set $X$, the order relation is assumed to be equality, i.e. $(X, =)$.

A class of graphs $\mathcal{C}$ is a *hereditary class* if it is closed under taking induced subgraphs, i.e. for $G \in \mathcal{C}$, if $H \subseteq_i G$, then $H \in \mathcal{C}$. The *hereditary closure* of a class of graphs $\mathcal{C}$ is the smallest hereditary class containing $\mathcal{C}$.

An *ordered graph* is a finite undirected graph $G = (V, E)$ equipped with a linear order $\leq_G$ on its vertices. The notion of induced subgraph relation extends to ordered graphs by requiring that the embedding $f \colon V_G \to V_H$ preserves the order, i.e., for all $u, v \in V_G$, $u \leq_G v$ if and only if $f(u) \leq_H f(v)$.

*Well-quasi-orderings.* A quasi-ordered set $(X, \leq)$ is a set $X$ with a reflexive and transitive relation $\leq$. A sequence $(x_i)_{n \in \mathbb{N}}$ of elements in $X$ is *good* if there exist $i < j$ such that $x_i \leq x_j$. A quasi-ordered set is *well-quasi-ordered* (*wqo*) if every infinite sequence is good. A *bad sequence* is an infinite sequence that is not good, and a sequence of pairwise incomparable elements is called an *antichain*.

Mael: an antichain is the same thing as bad sequence? How it is presented may be confusing.

As mentioned in the introduction, several notions of well-quasi-ordering can be defined for a given class of finite undirected graphs $\mathcal{C}$. Let us briefly state them: $\mathcal{C}$ is *well-quasi-ordered* if the class $(\mathcal{C}, \subseteq_i)$ is well-quasi-ordered, $\mathcal{C}$ is *$k$-well-quasi-ordered* for some $k \in \mathbb{N}$, if for every set $(X, =)$ of size $k$, the class $(\mathrm{Label}_X(\mathcal{C}), \subseteq_i)$ is well-quasi-ordered, and $\mathcal{C}$ is *$\forall$-well-quasi-ordered* if for every well-quasi-order $(X, \leq)$, the class $(\mathrm{Label}_X(\mathcal{C}), \subseteq_i)$ is well-quasi-ordered.

Finally, a class of graph is *orderably well-quasi-ordered* (resp. *orderably $k$-well-quasi-ordered*, resp. *orderably $\forall$-well-quasi-ordered*) if there is a way to equip each graph $G$ in $\mathcal{C}$ with a linear order $\leq_G$ on its vertices such that the resulting class of ordered graphs is well-quasi-ordered (resp. $k$-well-quasi-ordered, resp. $\forall$-well-quasi-ordered).

*Trees.* In this paper, *trees* are finite, binary, and ordered (have *left* and *right children*). They are understood as finite relational structures over the signature $\sigma = (\sqsubseteq, \preceq)$ where $\sqsubseteq$ is the *ancestor relation*, and $\preceq$ is the *sibling order* on the nodes of the tree, extended to all pairs of nodes as a topological order. In a tree $\mathcal{T}$, the *root* is denoted by $\mathsf{root}(\mathcal{T})$, with $\mathcal{T}$ being left implicit when clear from context. The set of *leaves* is denoted by $\mathsf{Leaves}(\mathcal{T})$. Given two nodes $x$ and $y$ in a tree $T$, their *least common ancestor* is denoted by $\mathsf{lca}(x, y)$. We will often use the *parent relation*, which state that $y$ is the unique immediate ancestor of $x$.

As in the case of graphs, we will consider vertex-labelled trees where labels are taken from a well-quasi-order $(X, \leq)$. We will also consider edge-labelled trees, typically using a finite set of labels $\Sigma$. The class of vertex-labelled and edge-labelled trees is denoted by $\mathsf{Trees}^X_\Sigma$. The sets $\Sigma$ or $X$ are omitted when unused (i.e., Trees denotes the class of unlabelled trees) Given two nodes $x, y$ such that $y$ is an *ancestor* of $x$, the notation $[x : y]_{\mathcal{T}}$ denotes the *word* in $\Sigma^*$ obtained by reading the labels of the edges on the path from $x$ to $y$ in $\mathcal{T}$, we can leave $\mathcal{T}$ implicit if clear from context.

*Bounded Clique-Width.* An MSO-*interpretation* $\mathcal{I}$ from $\mathsf{Trees}_\Sigma$ to finite undirected graphs is given by a tuple of MSO formulas that specify how to respectively: select a subset of the leaves of tree to be the vertices of the graph $\phi_{\mathsf{univ}}(x)$, define the edges of the graph $\phi_{\mathsf{edge}}(x, y)$, and define the domain of the interpretation $\phi_{\mathsf{dom}}$. An interpretation $\mathcal{I}$ is called *simple* if $\phi_{\mathsf{univ}}(x)$ and $\phi_{\mathsf{dom}}$ are tautologies.

*Example 2.1.* An MSO-interpretation $\mathcal{I}$ from Trees to finite undirected graphs that produces all paths can be defined as follows: $\phi_{\mathsf{dom}}$ is the formula that is true on linear trees (i.e., trees where every node has at most one non-leaf child), $\phi_{\mathsf{univ}}(x)$ is the formula that is true if and only if $x$ is a leaf and its sibling is not a leaf, $\phi_{\mathsf{edge}}(x, y)$ is the formula that is true if and only if are at distance exactly 3.

A class of graph has *bounded clique-width* if it is included in the image of some MSO-interpretation from finite trees to finite undirected graphs [8]. It has *bounded linear clique-width* if it is included in the image of some MSO-interpretation from linear trees (i.e., trees where every node has at most one non-leaf child) to finite undirected graphs. Equivalently, a class of graphs has bounded linear clique-width if it is included in the image of some simple MSO-interpretation from finite words to finite undirected graphs.

*Transductions from graphs to graphs.* Another notion of logical transformation that will be relevant in this paper is that of existential transduction and existential interpretation that were mentioned in Theorem 1.7 and Conjecture 1.3, and are defined in a similar way as MSO-interpretations.

An *existential interpretation* from (labelled) graphs to graphs is given by an existential first order formula $\phi_{\mathsf{univ}}(x)$ that selects the vertices of the interpreted graph, an existential first order formula $\phi_{\mathsf{edge}}(x, y)$ that define the edges of the interpreted graph, and an existential first order formula $\phi_{\mathsf{dom}}$ that selects the graphs of $C$ on which the interpretation is defined. The semantics is that for every graph $G \in C$ satisfying $\phi_{\mathsf{dom}}$, the interpreted graph $\mathcal{I}(G)$ has for vertex set the vertices of $G$ satisfying $\phi_{\mathsf{univ}}$, and has an edge between two vertices $u, v$ if and only if $G$ satisfies $\phi_{\mathsf{edge}}(u, v)$ (which is assumed to be symmetric and irreflexive on the selected vertices). A class $C$ *existentially interprets* a class $\mathcal{D}$ if there exists an existential interpretation $\mathcal{I}$ from $C$ to graphs such that $\mathcal{D} \subseteq \mathcal{I}(C)$. We say that $C$ *existentially transduces* $\mathcal{D}$ if there exists a finite labelling set $\Sigma$ such that $\mathsf{Label}_\Sigma(C)$ existentially interprets $\mathcal{D}$.

## 3 FROM INTERPRETATIONS TO NESTED TREES

As a first step towards proving Theorem 1.5, we are going to transform a simple MSO-interpretation from trees to graphs into a more combinatorial object that will make the interpretation *quantifier-free*. To that end, we will chain two standard transformations from automata theory: the first one is to transform an MSO-interpretation into a so-called monoid interpretation, and the second one is to factorise the trees using forward Ramseyan splits (nested trees) [5]. Finally, we will show that the natural ordering on nested trees (the gap embedding) is *almost* such that the resulting interpretation is order preserving from nested trees to graphs.

Let us fix for the rest of the section a finite alphabet $\Sigma$ and a simple MSO-interpretation $\mathcal{I}$ from $\mathsf{Trees}_\Sigma$ to finite undirected graphs defined by a formula $\varphi(x, y)$ that defines the edges of the graphs.

### 3.1 From Interpretation to Monoids

It follows from standard arguments that to a formula $\varphi(x, y)$ over $\mathsf{Trees}_\Sigma$ one can associate a finite monoid $M$ and a morphism $\mu\colon \Sigma^* \to M$ such that for any tree $T$ and any pair of leaves $x \preceq y$ in $T$, whether $T \models \varphi(x, y)$ is entirely determined by the values of $\mu([z : x]_{\mathcal{T}})$, $\mu([z : y]_{\mathcal{T}})$ and $\mu([\mathsf{root} : z]_{\mathcal{T}})$ where $z = \mathsf{lca}(x, y)$ is the least common ancestor of $x$ and $y$. We refer to the book on Tree Automata Techniques and Applications for a comprehensive overview of the connection between Monadic Second Order Logic on trees, and tree automata [7]. In the rest of the section, we will fix such a monoid $M$ and morphism $\mu$. To simplify notations, we will write $[\![x : y]\!]_{\mathcal{T}}$ for $\mu([x : y]_{\mathcal{T}})$, that is, the product of the labels of the edges on the path from $x$ to $y$ in $\mathcal{T}$.

As a consequence, one can collect in a subset $P \subseteq M^3$ all the triples $(a, b, c)$ corresponding to a satisfying assignments of the formula $\varphi(x, y)$. This is illustrated in Figure 2. We define a *monoid interpretation* from trees labelled over $M$ to finite undirected graphs to be a tuple $(\Sigma, \mu, M, P)$ where $\Sigma$ is a finite alphabet, $\mu\colon \Sigma^* \to M$ is a morphism to a finite monoid $M$, and $P \subseteq M^3$ is a subset of triples of elements of $M$. The interpretation works as follows: given a tree $\mathcal{T}$ whose edges are labelled by elements of $\Sigma$, the vertices of the interpreted graph are the leaves of $T$, and there is an edge between two leaves $x \preceq y$ if and only if the triple $([\![\mathsf{root} : \mathsf{lca}(x, y)]\!]_{\mathcal{T}}, [\![\mathsf{lca}(x, y) : x]\!]_{\mathcal{T}}, [\![\mathsf{lca}(x, y) : y]\!]_{\mathcal{T}})$ is in $P$.

Adding the function $[\![x : y]\!]_T$ to the signature of trees, and assuming that lca is also part of the signature, we can rewrite the formula $\varphi(x, y)$ as a quantifier-free formula as follows:

$$([\![\mathsf{root} : \mathsf{lca}(x, y)]\!]_{\mathcal{T}}, [\![\mathsf{lca}(x, y) : x]\!]_{\mathcal{T}}, [\![\mathsf{lca}(x, y) : y]\!]_{\mathcal{T}}) \in P \quad (1)$$

The main idea driving this paper is that understanding when a class of graphs is $\forall$-well-quasi-ordered can be reduced to finding a suitable well-quasi-order on trees such that the interpretation $\mathcal{I}$ is *order-preserving* from trees to graphs, leveraging the following standard Fact 3.1.

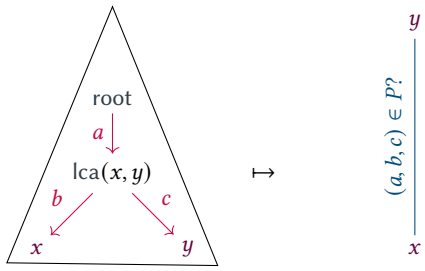



**Figure 2: The interpretation of a tree using a monoid $M$ and an accepting part $P \subseteq M^3$.**

*Fact 3.1.* Let $(X, \leq_X)$ and $(Y, \leq_Y)$ be two quasi-orders, and $f : X \to Y$ be a surjective order-preserving map. If $(X, \leq_X)$ is well-quasi-ordered, then $(Y, \leq_Y)$ is well-quasi-ordered.

Proof. Let $(y_i)_{i\in\mathbb{N}}$ be an infinite sequence of elements of $Y$. Since $f$ is surjective, there exists an infinite sequence $(x_i)_{i\in\mathbb{N}}$ of elements of $X$ such that for every $i$, $f(x_i) = y_i$. Since $(X, \leq_X)$ is well-quasi-ordered, there exists two indices $i < j$ such that $x_i \leq_X x_j$. Since $f$ is order-preserving, we have $y_i = f(x_i) \leq_Y f(x_j) = y_j$. □

One candidate ordering on trees is the *composition ordering* defined as follows. Given two trees $\mathcal{T}_1$ and $\mathcal{T}_2$ labelled over a monoid $M$, we say that $\mathcal{T}_1$ is less than $\mathcal{T}_2$ in the composition ordering, written $\mathcal{T}_1 \preceq_{\mathsf{cmp}} \mathcal{T}_2$, if there exists a map $h : \mathcal{T}_1 \to \mathcal{T}_2$ that maps leaves to leaves, respects the relation lca, $\preceq$, and the function $[\![\cdot : \cdot]\!]$. This ordering is naturally extended to node-labelled trees by asking that for every node $x$ in $\mathcal{T}_1$, the label of $x$ is less or equal than the label of $h(x)$ in $\mathcal{T}_2$. It is straightforward to check that the interpretation $\mathcal{I}$ is order-preserving from trees equipped with the composition ordering to graphs equipped with the induced subgraph ordering. Hence, to prove that the image of $\mathcal{I}$ is $\forall$-well-quasi-ordered, it suffices to prove that the class of trees is well-quasi-ordered under the composition ordering when labelled using a well-quasi-ordered set $(X, \leq_X)$.

Historically, this approach was used to prove that some classes of graphs are labelled-well-quasi-ordered, see for instance the results of [9, 12]. Unfortunately, the composition ordering on trees is often way stricter than the induced subgraph ordering. It was observed that the composition ordering on trees is well-quasi-ordered if and only if *for every* $P \subseteq M^3$, the class of graphs obtained from trees by considering only the triples in $P$ is labelled-well-quasi-ordered [17, Theorem 24]. To understand what happens for a precise choice of $P$, we need to have a finer understanding of the way trees can be decomposed.

## 3.2 Forward Ramseyan Splits

The combinatorial ingredient allowing us to further decompose the trees will come from an adaptation of the classical Simon Factorisation Theorem for semigroups [21], adapted to trees by Colcombet [5]. The rest of this section is devoted to explaining how this factorisation works, and how it can be used to define a more suitable ordering on trees.

A *split of height $N$* of a tree $\mathcal{T}$ is a mapping $\mathfrak{s}$ from the nodes of $T$ to $\{1, \ldots, N\}$. Given a split $\mathfrak{s}$ and two nodes $x \sqsubset_{\mathcal{T}} y$, we define

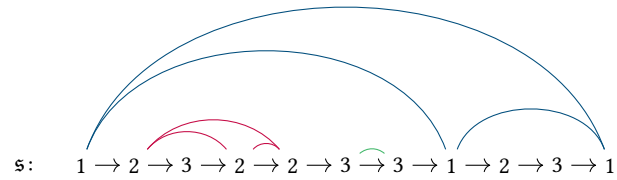



**Figure 3: A split $\mathfrak{s}$ on a branch of a tree. The arcs in blue connect 1-neighbours, the arcs in red connect 2-neighbours, and the arcs in green connect 3-neighbours.**

$\mathfrak{s}(x : y)$ to be the minimal value of $\mathfrak{s}(z)$ for $x \sqsubset_{\mathcal{T}} z \sqsubset_{\mathcal{T}} y$, and $N + 1$ otherwise. In particular, if $y$ is a child of $x$, then $\mathfrak{s}(x : y) = N + 1$.

Two elements $x \sqsubset_{\mathcal{T}} y$ such that $\mathfrak{s}(x) = \mathfrak{s}(y) = k$ are *$k$-neighbours* if $\mathfrak{s}(x : y) \geq k$. Being $k$-neighbours is an equivalence relation on the nodes of the tree having value $k$ in the split.

Mael: This is false, for example if a,b,c, are such that a is the lca of b and c; we can have a,b and a,c are k-nbr, but not b,c k-nbd. It should be defined branch-wise.

A split induces a hierarchical structure on the tree based on the $k$-neighbourhoods. We illustrate the situation on a *branch* of a tree in Figure 3, but in general $k$-neighbourhoods can span several branches of the tree. We will refer to the value of $\mathfrak{s}(x)$ as the *split depth* of the node $x$.

Mael: What is a class of $k$-neighbourhood ? I think the way you define it is not clear yet. You use it exclusively on branch, I think you should define it only on branch, but observe that on the whole tree, the equivalence classes intersect in a tree-like manner.

A split $\mathfrak{s}$ is *forward Ramseyan* if for every $k \in \{1, \ldots, N\}$ and every $x, y, x', y'$ in the same class of $k$-neighbourhood with $x \sqsubset_{\mathcal{T}} y$ and $x' \sqsubset_{\mathcal{T}} y'$, we have:

$$[\![x : y]\!]_{\mathcal{T}} = [\![x : y]\!]_{\mathcal{T}} \cdot [\![x' : y']\!]_{\mathcal{T}} \quad . \tag{2}$$

In particular, $[\![x : y]\!]_{\mathcal{T}}$ is always an *idempotent*: $[\![x : y]\!]_{\mathcal{T}} \cdot [\![x : y]\!]_{\mathcal{T}} = [\![x : y]\!]_{\mathcal{T}}$. However, note that $[\![x : y]\!]_{\mathcal{T}}$ and $[\![x' : y']\!]_{\mathcal{T}}$ may be different idempotents.

Let us illustrate how one can use forward Ramseyan splits to compute efficiently the value of $[\![x : y]\!]_{\mathcal{T}}$ for any pair of nodes $x \sqsubset_{\mathcal{T}} y$. We say that $x$ and $y$ are *independent at level $k$* if $\mathfrak{s}(x : y) = k$ and there exist three nodes $z_1, z_2, z_3$ such that $x \sqsubseteq_{\mathcal{T}} z_1 \sqsubset_{\mathcal{T}} z_2 \sqsubseteq_{\mathcal{T}} z_3 \sqsubseteq_{\mathcal{T}} y$ and $\mathfrak{s}(z_1) = \mathfrak{s}(z_2) = \mathfrak{s}(z_3) = k$, and $\mathfrak{s}(x : z_1) > k$, $\mathfrak{s}(z_1 : z_2) > k$, and $\mathfrak{s}(z_3 : y) > k$. The idea being that $x$ and $y$ are *separated* by at least one $k$-neighbourhood. In the case of independence at level $k$, we can use the property of forward Ramseyan splits to compute $[\![x : y]\!]_{\mathcal{T}}$ without ever looking at the section between $z_2$ and $z_3$, leading to a fast (and first-order definable) computation. We refer to [5, Lemma 3] for more details on this computation.

The main theorem of [5] is that for every finite monoid $M$, there exists a finite depth $N$ such that for every tree $\mathcal{T}$ edge-labelled using $M$, there exists a forward Ramseyan split of height $N$ for $\mathcal{T}$. We can therefore assume that our trees are always equipped with a forward Ramseyan split of height $N$.

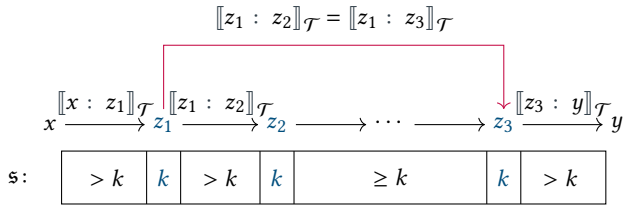



**Figure 4: Fast computation of the value $[\![x : y]\!]_{\mathcal{T}}$ provided a forward Ramseyan split, using the fact that $x$ and $y$ are independent at level $k$.**

Let us now state the main remark that will guide the rest of this paper: given a branch of a tree equipped with a forward Ramseyan split as the one illustrated in Figure 4, the value of $[\![x : y]\!]_{\mathcal{T}}$ does not change when inserting new nodes in the section between $z_2$ and $z_3$ as long as these new nodes are assigned a value of the split bigger than or equal to $k$, and the resulting tree is still forward Ramseyan. A fortiori, whether there is an edge between two leaves of $x$ and $y$ in the resulting graph is not influenced by such insertions. This will guide our definition of a suitable ordering on trees in the upcoming Section 3.3.

## 3.3 Marked Nested Trees

In this section, we will show how the notion of forward Ramseyan split gets us closer to defining a well-quasi-order on trees that is suitable for our purpose. A first step is to notice that there already exists a notion of embedding that somehow respects forward Ramseyan splits, called gap-embeddings and that endows trees with a well-quasi-order [11].

*Definition 3.2 ([11, Definition 3.3]).* A *gap-embedding* between trees $\mathcal{T}_1, \mathcal{T}_2$ endowed with forward Ramseyan splits $\mathfrak{s}_1, \mathfrak{s}_2$ is a mapping $h\colon \mathcal{T}_1 \to \mathcal{T}_2$ that

(1) is a tree embedding: respects the ancestor relation $\sqsubseteq$, least common ancestors lca, and sibling ordering $\preceq$,
(2) satisfies the root gap property: $\mathfrak{s}_2(r_2 : h(r_1)) \geq \mathfrak{s}_1(r_1)$ where $r_1, r_2$ are the roots of $\mathcal{T}_1, \mathcal{T}_2$,
(3) satisfies the edge gap property: for every edge $x \sqsubset_{\mathcal{T}} y$ in $\mathcal{T}_1$, $\mathfrak{s}_2(h(x) : h(y)) \geq \mathfrak{s}_1(y)$.

If the trees are node-labelled by a set $(X, \leq_X)$, we further require that for every node $x$ in $\mathcal{T}_1$, the label of $x$ is less than or equal to the label of $h(x)$ in $\mathcal{T}_2$.

In the following Fact 3.3 we illustrate why gap-embeddings are well-suited for our purpose: they respect the gaps induced by forward Ramseyan splits.

*Fact 3.3.* Let $\mathcal{T}_1, \mathcal{T}_2$ be two trees and $\mathfrak{s}_1, \mathfrak{s}_2$ be two forward Ramseyan splits of height $N$ of $\mathcal{T}_1, \mathcal{T}_2$ Assume that $h\colon \mathcal{T}_1 \to \mathcal{T}_2$ is a gap-embedding from $(\mathcal{T}_1, \mathfrak{s}_1)$ to $(\mathcal{T}_2, \mathfrak{s}_2)$. Then, for every $u, v$ in $\mathcal{T}_1$, if $\mathfrak{s}_1(u : v) > k$ and $\mathfrak{s}_1(v) = k$ for some $k \in \{1, \ldots, N\}$, then $\mathfrak{s}_2(h(u) : h(v)) \geq k$.

Proof. This is proven by induction on the length of the path from $u$ to $v$ in $\mathcal{T}_1$. The base case is exactly ?? of the definition of gap-embedding. □

One other key property of the gap-embedding ordering is that it is a well-quasi-ordering.

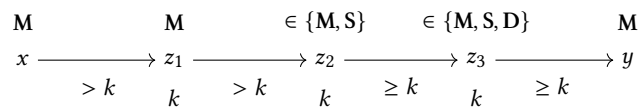
Theorem 3.4 ([11, Theorem 3.1, Main Theorem]). *The class of trees equipped with forward Ramseyan splits is well-quasi-ordered under the gap-embedding ordering, even under the assumption that nodes are labelled with a well-quasi-ordered set $(X, \leq_X)$.*

We will now devise a suitable modification of the gap-embedding ordering that will ensure that the interpretation $\mathcal{I}$ is order-preserving from trees to graphs. In particular, this modification will take into account the edge-labelling of the trees using the monoid $M$, that are not considered in the gap-embedding ordering.

*Definition 3.5.* A *marked nested tree* is a tuple $(\mathcal{T}, \mathfrak{s}, \rho)$ where $\mathcal{T}$ is a tree whose edges are labelled over a finite monoid $M$, $\mathfrak{s}$ is a forward Ramseyan split of height $N$ of $\mathcal{T}$, and $\rho$ is a function from the nodes of $\mathcal{T}$ to $\{\mathbf{M}, \mathbf{S}, \mathbf{D}\}$, respectively called *marked*, *separating*, and *dummy* nodes.

Mael: I think there is something wrong with this definition. "that for every $x \sqsubset_{\mathcal{T}} z_1 \sqsubset_{\mathcal{T}} y$ such that $\mathfrak{s}(x : z_1) > k$, $\mathfrak{s}(z_1) = k$" there is only one such $z_1$, Moreover, you assume $\rho(z_1) = \mathbf{M}$ afterward.

A marked nested tree $(\mathcal{T}, \mathfrak{s}, \rho)$ is *well-marked* if the marked nodes contain the root of $\mathcal{T}$, are closed under the least common ancestor operation, and furthermore satisfy that for every $x \sqsubset_{\mathcal{T}} z_1 \sqsubset_{\mathcal{T}} y$ such that $\mathfrak{s}(x : z_1) > k$, $\mathfrak{s}(z_1) = k$, and $\mathfrak{s}(z_1 : y) \geq k$ for some $k \in \{1, \ldots, N\}$, if $x$ and $y$ are marked nodes and $z_1$ is not a marked node, then there exists $z_2, z_3$ such that $x \sqsubset_{\mathcal{T}} z_1 \sqsubset_{\mathcal{T}} z_2 \sqsubseteq_{\mathcal{T}} z_3 \sqsubseteq_{\mathcal{T}} y$, $\mathfrak{s}(z_1) = \mathfrak{s}(z_2) = \mathfrak{s}(z_3) = k$, $\mathfrak{s}(x : z_1) > k$, $\mathfrak{s}(z_1 : z_2) > k$, $\mathfrak{s}(z_2 : z_3) \geq k$, and $\mathfrak{s}(z_3 : y) \geq k$, with $\rho(z_1) = \mathbf{M}$, and $\rho(z_2) \in \{\mathbf{M}, \mathbf{S}\}$. A graphical illustration of the condition is provided below:




*Definition 3.6.* Given two marked nested trees $(\mathcal{T}_1, \mathfrak{s}_1, \rho_1)$ and $(\mathcal{T}_2, \mathfrak{s}_2, \rho_2)$, we say that $(\mathcal{T}_1, \mathfrak{s}_1, \rho_1)$ *gap-embeds* into $(\mathcal{T}_2, \mathfrak{s}_2, \rho_2)$, written $(\mathcal{T}_1, \mathfrak{s}_1, \rho_1) \leq_{\text{gap}} (\mathcal{T}_2, \mathfrak{s}_2, \rho_2)$, if there exists a mapping $h\colon \mathcal{T}_1 \to \mathcal{T}_2$ that

★ is a gap embedding from $(\mathcal{T}_1, \mathfrak{s}_1)$ to $(\mathcal{T}_2, \mathfrak{s}_2)$,
(4) sends the root of $\mathcal{T}_1$ to the root of $\mathcal{T}_2$,
(5) maps leaves to leaves,
(6) respects the marking: $\rho_1 = \rho_2 \circ h$,
(7) respects local products: if $y$ is the left (resp. right) child of $x$ in $\mathcal{T}_1$, and $y'$ is the left (resp. right) child of $h(x)$ in $\mathcal{T}_2$, then $[\![x : y]\!]_{\mathcal{T}_1} = [\![h(x) : y']\!]_{\mathcal{T}_2}$.
(8) respects neighbourhood products: for every $k \in \{1, \ldots, N\}$, and every node $x \in \mathcal{T}_1$, if $z$ is the closest ancestor of $x$ such that $\mathfrak{s}_1(z) = k$ and $z'$ is the closest ancestor of $h(x)$ such that $\mathfrak{s}_2(z') = k$, then $[\![z : x]\!]_{\mathcal{T}_1} = [\![z' : h(x)]\!]_{\mathcal{T}_2}$.
(9) is gluing on non-dummy nodes: if $\rho_1(y) \in \{\mathbf{M}, \mathbf{S}\}$, and $x$ is the parent of $y$ in $\mathcal{T}_1$, then $h(x)$ is the parent of $h(y)$ in $\mathcal{T}_2$.

Unlike the gap-embedding ordering of Theorem 3.4, the marked gap-embedding ordering of Definition 3.6 is not a well-quasi-order because of the gluing condition of Item 9. We create an infinite

antichain by considering the infinite sequence of trees $(\mathcal{T}_n, \mathfrak{s}_n, \rho_n)$ where $\mathcal{T}_n$ is a branch of length $n+1$, $\mathfrak{s}_n$ assigns the value 1 to every node of $\mathcal{T}_n$, and $\rho_n$ marks all the nodes of $\mathcal{T}_n$ with $\mathbf{M}$. It is straightforward to check that for every $n < m$, $(\mathcal{T}_n, \mathfrak{s}_n, \rho_n)$ does not gap-embed into $(\mathcal{T}_m, \mathfrak{s}_m, \rho_m)$.

To recover the well-quasi-order property, we will restrict our attention to trees that make a limited use of the gluing property. A $L$-bounded marked nested tree is a marked nested tree such that any path of non dummy nodes has length at most $L$. As we will see in the following Theorem 3.7, the class of $L$-bounded well-marked nested trees is well-quasi-ordered under the marked gap-embedding ordering when labelled over a well-quasi-ordered set.

Theorem 3.7. *For every finite $L$, the class of $L$-bounded well-marked nested trees labelled over a well-quasi-ordered set $(X, \leq_X)$ is well-quasi-ordered under the gap-embedding ordering.*

Proof. We provide a function $f$ that maps $L$-bounded well-marked nested trees into nested trees labelled over a well-quasi-ordered set, such that if there exists a gap-embedding between $f(\mathcal{T}_1, \mathfrak{s}_1, \rho_1)$ and $f(\mathcal{T}_2, \mathfrak{s}_2, \rho_2)$, then $\mathcal{T}_1 \leq_{\text{gap}} \mathcal{T}_2$ as marked nested trees. Since we know that the class of nested trees labelled over a well-quasi-ordered set is well-quasi-ordered under the gap-embedding ordering by Theorem 3.4, we will immediately conclude that the class of $L$-bounded well-marked nested trees is also well-quasi-ordered under the marked gap-embedding ordering. One can encode Items 4 to 8 of Definition 3.6 by adding suitable labels to the nodes of the trees. Formally, we define a new labelling of the nodes of a tree $\mathcal{T}$ as follows: we add to each node $x$ of $\mathcal{T}$ the label $[\![z : x]\!]_{\mathcal{T}}$ for every $k \in \{1, \dots, N\}$, where $z$ is the least ancestor of $x$ such that $\mathfrak{s}(z) = k$, or a special symbol $\perp$ if no such ancestor exists. Furthermore, if $y$ is the immediate left (resp. right) child of $x$ in $\mathcal{T}$, we also add the label $[\![x : y]\!]_{\mathcal{T}}$ to $x$, or the special symbol $\perp$ if no such child exists. We also distinguish the root by adding a special label root, and we add a special label leaf to every leaf. Finally, we also add the label $\rho(x)$ to $x$. Since $M$ is finite, the new labelling is still over a well-quasi-ordered set.

It remains to ensure that the mapping $h$ is gluing on non-dummy nodes. To that end, we will modify the split $\mathfrak{s}$ into a new split $\mathfrak{s}'$ as follows: for every non-dummy node $x$, we have that $x$ is the $i$th element in a maximal path of non-dummy nodes $x_1 \sqsubset_{\mathcal{T}} x_2 \sqsubset_{\mathcal{T}} \cdots \sqsubset_{\mathcal{T}} x_n$ that contains $x$. Because the tree is $L$-bounded, we have $n \leq L$. We set $\mathfrak{s}'(x) = N + 1 + i$. For a dummy node $y$, we set $\mathfrak{s}'(y) = \mathfrak{s}(y)$.

Note that any gap-embedding $h$ from $(\mathcal{T}_1, \mathfrak{s}'_1)$ to $(\mathcal{T}_2, \mathfrak{s}'_2)$ is necessarily gluing on non-dummy nodes: assume that $\rho_1(y) \in \{\mathbf{M}, \mathbf{S}\}$, and let $x$ be the parent of $y$ in $\mathcal{T}_1$. Since $\mathfrak{s}'_1(y) = N + 1 + i$, we know that $\mathfrak{s}'_2(h(x) : h(y)) \geq \mathfrak{s}'_1(y) = N + 1 + i$. But this means that between $h(x)$ and $h(y)$ there are only nodes with split depth greater or equal than $N + 1 + i$. If $i = 1$ (i.e., $y$ is the first node of a maximal path of non-dummy nodes), then there is no such node, and $h(x)$ is the parent of $h(y)$. Otherwise, $\mathfrak{s}'_1(x) = N + i > N$, and therefore the path from $h(x)$ to $h(y)$ contains only non dummy nodes. Since the split values have been chosen to be strictly increasing inside maximal paths of non-dummy nodes, there is no such node, and $h(x)$ is the parent of $h(y)$. □

Lemma 3.8. *Let $(\mathcal{T}_1, \mathfrak{s}_1, \rho_1)$ and $(\mathcal{T}_2, \mathfrak{s}_2, \rho_2)$ be two marked nested trees such that $(\mathcal{T}_1, \mathfrak{s}_1, \rho_1) \leq_{\text{gap}} (\mathcal{T}_2, \mathfrak{s}_2, \rho_2)$ via some mapping $h$. For every pair of marked nodes $x \sqsubset_{\mathcal{T}_1} y$ in $\mathcal{T}_1$, we have $[\![x : y]\!]_{\mathcal{T}_1} = [\![h(x) : h(y)]\!]_{\mathcal{T}_2}$.*

Proof. We proceed by induction on the value of $\mathfrak{s}_1(x : y)$. If $\mathfrak{s}_1(x : y) = N + 1$, then $y$ is a child of $x$. Because $y$ is a marked node, then by Item 9 we know that $h(y)$ is a child of $h(x)$. Therefore, by Item 7, we have $[\![x : y]\!]_{\mathcal{T}_1} = [\![h(x) : h(y)]\!]_{\mathcal{T}_2}$.

Otherwise, let $k = \mathfrak{s}_1(x : y) \leq N$. Since $(\mathcal{T}_1, \mathfrak{s}_1, \rho_1)$ is well-marked, there exists $z_1, z_2, z_3$ such that $x \sqsubset_{\mathcal{T}_1} z_1 \sqsubset_{\mathcal{T}_1} z_2 \sqsubseteq_{\mathcal{T}_1} z_3 \sqsubseteq_{\mathcal{T}_1} y$, $\mathfrak{s}_1(z_1) = \mathfrak{s}_1(z_2) = \mathfrak{s}_1(z_3) = k$, and $\mathfrak{s}_1(x : z_1) > k$, $\mathfrak{s}_1(z_1 : z_2) > k$, $\mathfrak{s}_1(z_2 : z_3) \geq k$, and $\mathfrak{s}_1(z_3 : y) > k$, with $\rho_1(z_1) = \mathbf{M}$, and $\rho_1(z_2) \in \{\mathbf{M}, \mathbf{S}\}$. This is depicted in the following figure:

$$x \xrightarrow[>k]{[\![x : z_1]\!]_{\mathcal{T}_1}} \underset{k}{z_1} \xrightarrow[>k]{[\![z_1 : z_2]\!]_{\mathcal{T}_1}} \underset{k}{z_2} \xrightarrow[\geq k]{[\![z_2 : z_3]\!]_{\mathcal{T}_1}} \underset{k}{z_3} \xrightarrow[>k]{[\![z_3 : y]\!]_{\mathcal{T}_1}} y$$

By applying the gap-embedding $h$, we have the following configuration inside $\mathcal{T}_2$, where we used the notations $a = [\![h(x) : h(z_1)]\!]_{\mathcal{T}_2}$, $b = [\![h(z_1) : h(z_2)]\!]_{\mathcal{T}_2}$, $c = [\![h(z_2) : h(z_3)]\!]_{\mathcal{T}_2}$, and $d = [\![h(z_3) : h(y)]\!]_{\mathcal{T}_2}$.

$$h(x) \xrightarrow[>k]{a} \underset{k}{h(z_1)} \xrightarrow[>k]{b} \underset{k}{h(z_2)} \xrightarrow[\geq k]{c} \underset{k}{h(z_3)} \xrightarrow[>k]{d} h(y)$$

Let us briefly argue the inequalities on the splits on the above paths. The non-strict inequalities follow from Fact 3.3. Furthermore, if $v$ is a marked node or a separating node, then we can strengthen the inequality to a strict one, i.e., $\mathfrak{s}(h(u) : h(v)) > k$ even when $\mathfrak{s}(v) \neq k$. This is because of Item 9, which ensures that the immediate parent of $h(v)$ is the image of the immediate parent of $v$, which has a split value strictly greater than $k$, and we conclude by the previous argument.[2]

Using these facts we conclude that:

$$\begin{aligned}
\mathfrak{s}_2(h(x) : h(z_1)) &> k && \text{because } z_1 \text{ is a marked,} \\
\mathfrak{s}_2(h(z_1) : h(z_2)) &> k && \text{because } z_2 \text{ is a marked or separating,} \\
\mathfrak{s}_2(h(z_2) : h(z_3)) &\geq k && \text{by the gap property,} \\
\mathfrak{s}_2(h(z_3) : h(y)) &> k && \text{because } y \text{ is a marked,}
\end{aligned}$$

Now, we can compute the values of $a$, $b$, and $d$:

$$\begin{aligned}
[\![x : z_1]\!]_{\mathcal{T}_1} &= a && \text{by induction hypothesis,} \\
[\![z_1 : z_2]\!]_{\mathcal{T}_1} &= b && \text{by Item 8,} \\
[\![z_3 : y]\!]_{\mathcal{T}_1} &= d && \text{by Item 8.}
\end{aligned}$$

[2]The case where there are no intermediate nodes between $u$ and $v$ happens if $v$ is the immediate child of $u$, in which case the property is given directly by the gluing property.

Finally, because $h(x)$ and $h(y)$ are independent at level $k$, and so are $x$ and $y$, we have by Equation (2) that

$$\begin{aligned}
[\![x : y]\!]_{\mathcal{T}_1} &= [\![x : z_1]\!]_{\mathcal{T}_1} \cdot [\![z_1 : z_2]\!]_{\mathcal{T}_1} \cdot [\![z_2 : z_3]\!]_{\mathcal{T}_1} \cdot [\![z_3 : y]\!]_{\mathcal{T}_1} \\
&= [\![x : z_1]\!]_{\mathcal{T}_1} \cdot [\![z_1 : z_2]\!]_{\mathcal{T}_1} \cdot [\![z_3 : y]\!]_{\mathcal{T}_1} \\
&= a \cdot b \cdot d \\
&= a \cdot b \cdot c \cdot d \\
&= [\![h(x) : h(y)]\!]_{\mathcal{T}_2} \quad . \qquad \square
\end{aligned}$$

Corollary 3.9. *Let $(\mathcal{T}_1, \mathfrak{s}_1, \rho_1)$ and $(\mathcal{T}_2, \mathfrak{s}_2, \rho_2)$ be two well-marked nested trees such that $(\mathcal{T}_1, \mathfrak{s}_1, \rho_1) \leq_{\text{gap}} (\mathcal{T}_2, \mathfrak{s}_2, \rho_2)$. Then, the graph interpreted from $(\mathcal{T}_1, \mathfrak{s}_1, \rho_1)$ is an induced subgraph of the graph interpreted from $(\mathcal{T}_2, \mathfrak{s}_2, \rho_2)$ when considering only marked leaves as vertices.*

Proof. Follows directly from Lemma 3.8 and the definition of the monoid interpretation using the set $P \subseteq M^3$ (see Equation (1)): edges between two marked leaves $x$ and $y$ only depend on the values of $[\![\text{root} : \text{lca}(x, y)]\!]_{\mathcal{T}}$, $[\![\text{lca}(x, y) : x]\!]_{\mathcal{T}}$, and $[\![\text{lca}(x, y) : y]\!]_{\mathcal{T}}$, all of which are marked nodes. $\square$

## 4 BAD PATTERNS IN IMAGES OF INTERPRETATIONS

Mael: You need to mention somewhere in this section that you fix an interpretation $\mathcal{I}$ and an alphabet $\Sigma$. (this is important because for exemple in Lemma 4.5 it is not clear what is the image of $\mathcal{I}$)

In this section, we will isolate combinatorial obstructions to being 2-well-quasi-ordered in classes of graphs that are images of monoid interpretations from trees. We will take the same notations as in Section 3 regarding forward Ramseyan splits and monoid interpretations. The key idea of this section is to start from a tree $\mathcal{T}$ equipped with a forward Ramseyan split $\mathfrak{s}$, and try to understand under which conditions one can turn this tree into a $L$-bounded marked nested tree representing the same graph, allowing us to leverage Theorem 3.7 to conclude that the class of graphs is $\forall$-well-quasi-ordered.

*Definition 4.1.* Let $\mathcal{T}$ be a tree and $\mathfrak{s}$ be a forward Ramseyan split of height $N$. A *bough of level $k$* in $\mathcal{T}$ is defined from a part of a $k$-neighbourhood as follows: Consider a sequence $b_0, b_1, \ldots, b_n$ of nodes of level $k$ on a branch of $\mathcal{T}$ forming a $k$-neighbourhood. Elements of the *bough* $B$ are all nodes $x$ in $\mathcal{T}$ such that $b_0 \sqsubseteq x$, and $\neg(b_n \sqsubseteq x)$, plus $b_n$ itself.

We say that the *dimension* of the bough $B$ is $n$, and that $b_0, b_1, \ldots, b_n$ form the *backbone* of $B$.

Mael: Reformulation suggestion. Question: does $b_0, b_1, \ldots, b_n$ have to be maximal? or maximal between $b_0$ and $b_n$ (i.e. no vertices not in this sequence in the same k-neighborhood)?

*Definition 4.2.* Let $\mathcal{T}$ be a tree and $\mathfrak{s}$ be a forward Ramseyan split of height $N$. Let $b_0, b_1, \ldots, b_n$ be any sequence of nodes of level $k$ on a branch of $\mathcal{T}$ forming a $k$-neighbourhood. A *bough of level $k$* is the subtree of $\mathcal{T}$ induced by all nodes $x$ such that $b_0 \sqsubseteq x$, and $\neg(b_n \sqsubseteq x)$, plus $b_n$ itself.

We say that the *dimension* of the bough $B$ is $n$, and that $b_0, b_1, \ldots, b_n$ form the *backbone* of $B$.

Let $B$ be a bough of level $k$ in $\mathcal{T}$ and let $b_0, b_1, \ldots, b_n$ be its backbone. Given indices $i < j$, we define the $\mathcal{T}_{b_i:b_j}$ as the set of nodes $x$ in $\mathcal{T}$ such that $b_i \sqsubseteq x$ and $\neg(b_j \sqsubseteq x)$. When $j = i + 1$, this set of nodes is called a *block*. We also refer to Figure 5 for an illustration of the resulting partition of the tree $\mathcal{T}$ with respect to a given bough $B$.

Mael: Alternative definition: For every leaf $x \in \mathcal{T}_{b_0:b_n}$, we define $\mathsf{B}_t(x) \triangleq \text{lca}(x, b_n)$. Moreover we define $\mathsf{B}_l(x)$ (resp. $\mathsf{B}_r(x)$) to be the maximal (resp. minimal) element of the backbone $b_0, b_1, \ldots, b_n$ such that $\mathsf{B}_l(x) \sqsubseteq \mathsf{B}_t(x) \sqsubseteq \mathsf{B}_r(x)$.

For every leaf $x \in \mathcal{T}_{b_0:b_n}$ where $n$ is the dimension of the bough $B$, we define $\mathsf{B}_t(x)$ to be the closest ancestor of $x$ such that $b_0 \sqsubseteq \mathsf{B}_t(x) \sqsubset b_n$, that is, the closest ancestor of $x$ on the branch going through the nodes $b_0, b_1, \ldots, b_n$. Similarly, we define $\mathsf{B}_l(x)$ and $\mathsf{B}_r(x)$ to be such that $\mathsf{B}_l(x)$ and $\mathsf{B}_r(x)$ belong to $\{b_0, b_1, \ldots, b_n\}$, and are the closest ones such that $\mathsf{B}_l(x) \sqsubseteq \mathsf{B}_t(x) \sqsubseteq \mathsf{B}_r(x)$. To every leaf $x$ of $B(\mathcal{T})$, Using these reference points, we define the following monoid elements associated to a leaf $x$ in a block of $B$:

$$\begin{aligned}
\mathcal{B}_t(x) &\triangleq [\![\mathsf{B}_t(x) : x]\!]_{\mathcal{T}}, \\
\mathcal{B}_l(x) &\triangleq [\![\mathsf{B}_l(x) : \mathsf{B}_t(x)]\!]_{\mathcal{T}}, \\
\mathcal{B}_r(x) &\triangleq [\![\mathsf{B}_t(x) : \mathsf{B}_r(x)]\!]_{\mathcal{T}}, \\
\mathcal{B}_{\text{root}}(x) &\triangleq [\![\text{root} : \mathsf{B}_l(x)]\!]_{\mathcal{T}}.
\end{aligned}$$

We call this tuple the *bough type* of the leaf $x$ with respect to the bough $B$. We again refer to Figure 5 for an illustration of the type of a leaf with respect to a given bough $B$. There are only finitely many possible bough types because $M$ is finite.

In the rest of the paper, we will ofter rely on some tree surgery operation consisting in replacing a bough in a tree by some other bough. Given a bough $B$ in a tree $\mathcal{T}$ defined using a $k$-neighbourhood $b_0, b_1, \ldots, b_n$, we write $\mathcal{T} = C[B]$ to denote that $\mathcal{T}$ can be obtained by plugging the bough $B$ into a *context* $C[\square]$. Formally, a *context* $C[\square]$ is given by a tree $\mathcal{T}_{\text{root}}$ with a distinguished leaf $\square$, together with two trees $\mathcal{T}_{\text{left}}$ and $\mathcal{T}_{\text{right}}$, and two monoid elements $m_{\text{left}}$ and $m_{\text{right}}$. These three elements are represented as dashed lines in Figure 5. The tree $C[B]$ is then obtained as follows: one first attaches $\mathcal{T}_{\text{left}}$ as the left subtree of $b_n$ with an edge labelled $m_{\text{left}}$, where $b_n$ is the last element of the $k$-neighbourhood defining $B$, then one attaches $\mathcal{T}_{\text{right}}$ as the right subtree of $b_n$ with an edge labelled $m_{\text{right}}$, and finally, one replaces the distinguished leaf $\square$ in $\mathcal{T}_{\text{root}}$ by the node $b_0$.

Mael: Suggestion: define leaves of $C[\square]$ to be the leaves of the context minus $\square$. This can be helpful when you want to talk about the subgraph defined by the leaves outside of $B$. (some following reformulation suggestion assume this is defined) (you actually use this definition in the following fact)

Mael: Define "start with the same idpt value" ($[\![b_0 : b_1]\!]_B$ ?). Also you sometime say "idempotent labelling the bough", this is the same and you should define it

Two boughs $B$ and $B'$ are *compatible* if they start with the same idempotent value, and if for every tree (with a split) $\mathcal{T} = C[B]$ that is forward Ramseyan, the tree $C[B']$ remains forward Ramseyan.

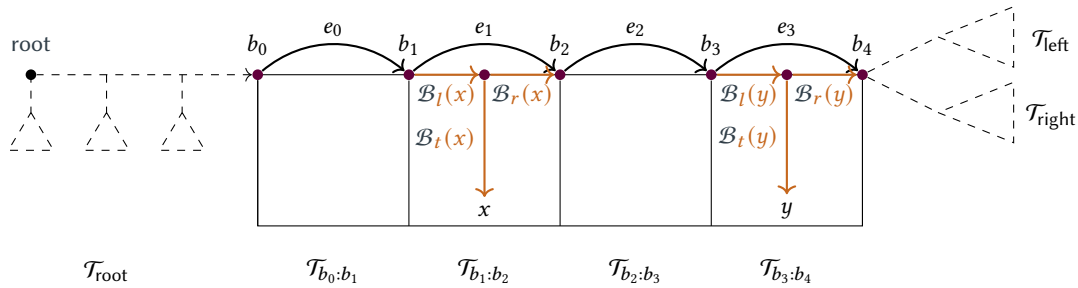



**Figure 5: Partitionning a branch of a tree $\mathcal{T}$ using a bough $B$. Here, the nodes $b_i$ for $0 \leq i \leq 4$ are $k$-neighbours forming the backbone $B$. The letter $e_i$ denotes idempotent monoid element $[\![b_i : b_{i+1}]\!]_{\mathcal{T}}$. In dashed, we represented the context $C[\square]$ such that $\mathcal{T} = C[B]$. To compute the presence of an edge between $x$ and $y$ in the resulting graph, it is sufficient to know the values of $[\![b_2 : b_3]\!]_{\mathcal{T}}$, and the respective bough types of $x$ and $y$ in their respective bough blocks $\mathcal{T}_{b_1:b_2}$ and $\mathcal{T}_{b_3:b_4}$.**

**Mael: I added some details on the figure 'on the root, and added the trees of the context, maybe you can even add labels mleft and mright on their edges. Moreover I think it would be better if we put $B_t, B_l, B_r$ instead of $\mathcal{B}_t(x)$ and co.**

Mael: Ptet tu peux essayer de rappeler ce que ça veut dire de rester forward ramseyan ici.

The following Fact 4.3 states that one can safely perform such a replacement without modifying the part of the graph represented by the tree $\mathcal{T}$ outside of the bough. This is because to determine the presence of edges between leaves outside of the bough, one only needs to know the value of the idempotent element labelling the bough, which does not change when replacing $B$ by a compatible bough $H$.

Mael: reformulation: safely perform such a replacement without modifying the subgraph defined by the leaves of $C[\square]$. This is because to determine the presence of edges between leaves of $C[\square]$, one only needs to know the value of the idempotent element labelling the bough, which does not change when replacing $B$ by a compatible bough $H$.

*Fact 4.3.* Let $\mathcal{T} = C[B]$ be a tree with a bough $B$ of level $k$, and let $H$ be a bough that is compatible with $B$. Let $\mathcal{T}' = C[H]$ be the tree obtained by replacing $B$ by $H$ in $\mathcal{T}$. Then, the subraph of $\mathcal{I}(\mathcal{T})$ induced by the leaves in the context $C[\square]$ equals the subgraph of $\mathcal{I}(\mathcal{T}')$ induced by the leaves in the context $C[\square]$.

*Definition 4.4.* Let $B$ be a bough of level $k$ in a tree $\mathcal{T}$. We say that $B$ is a *good bough* if, there exists a compatible bough $H$ of level $k$, and a map $h\colon \mathcal{I}(C[B]) \to \mathcal{I}(C[H])$ such that:

(1) $h$ is an embedding of graphs,
(2) $h$ is the identity map on leaves belonging $C[\square]$,
(3) there exists a pair of consecutive blocks in $H$ such that all their leaves are left untouched by $h$.

Mael: Why need two blocks ? this is problematic with the definition of perfect bough (as perfect is not anymore a restriction of good boughs)

A *bad bough* is a bough that is not a good bough.

We claim that the existence of bad boughs of arbitrarily large dimension is the only obstruction to being 2-well-quasi-ordered.

Mael: You should put an enumerate here between the equivalent notions, and then observe that 2->1, and say that in Section 4.1 you will show 3->2 and in Section 4.2 show that 1->3.

Lemma 4.5. *The image of $\mathcal{I}$ is 2-well-quasi-ordered if and only if the image of $\mathcal{I}$ is $\forall$-well-quasi-ordered if and only if there exists a bound on the dimension of bad boughs.*

The proof of Lemma 4.5 is split into two parts, one will leverage the bound on the dimension of bad boughs to construct a suitable tree representation of the graphs proving that they are $\forall$-well-quasi-ordering (Section 4.1). On the other hand, we will show that the existence of bad boughs of arbitrarily large dimension allows to construct an infinite two-labelled antichain in the image of $\mathcal{I}$ (Section 4.2).

## 4.1 Using the Gap Embedding

Mael: You need to explain why you want to insert dummy nodes

If there is an upper bound $N_0$ on the dimension of bad boughs, we can use Fact 4.3 to upgrade our trees in the following way: given a tree $\mathcal{T}$, we will insert dummy nodes in every bough of level $k$ that has dimension greater than $N_0$. The resulting tree $\mathcal{T}'$ will be a marked nested tree as defined in Section 3, with three kinds of nodes: marked nodes, that correspond to nodes of $\mathcal{T}$, dummy nodes that were inserted during this procedure, and separating nodes, that are the last nodes of the $k$-neighbourhoods defining the boughs we manipulated. The following Lemma 4.6 shows that this construction produces well-marked nested trees that are $L$-bounded, allowing us to apply Theorem 3.7.

Lemma 4.6. *Assume that there exists a bound $N_0$ on the dimension of bad boughs. Then, for every tree $\mathcal{T}$ equipped with a forward Ramseyan split $\mathfrak{s}$, one can effectively construct a well-marked nested tree $(\mathcal{T}', \mathfrak{s}', \rho)$ that is $L$-bounded for some $L \in \mathbb{N}$, and such that the restriction of $\mathcal{I}(\mathcal{T}')$ to marked leaves is isomorphic to $\mathcal{I}(\mathcal{T})$.*

Proof. We proceed iteratively starting from the tree $\mathcal{T}$ where every node is set as a marked node. A rewrite step is as follows: given a bough backbone $b_0, b_1, \ldots, b_n$ defining a bough $B$, where all $b_i$ are marked nodes, and $n = N_0 + 1$, we know that $B$ is a good bough by assumption. As a consequence, there exists a compatible bough $H$ of same level as $B$ that can replace $B$ in $\mathcal{T}$ by definition of good boughs. The resulting tree $\mathcal{T}'$ is still forward Ramseyan by definition of compatible boughs, and the graph induced by leaves outside of $B$ remains unchanged by Fact 4.3. Let us now describe the marking on $\mathcal{T}'$: leaves of $B$ that were marked remain marked in $H$, untouched leaves of $H$ become dummy, and if $H_{h_i:h_{i+1}}$ and $H_{h_{i+1}:h_{i+2}}$ are the two consecutive bough blocks of $H$ left untouched by the embedding witnessing that $B$ is a good bough, then we mark $h_i$ as a separating node, and all other nodes of these two bough blocks as dummy nodes. Then, we complete the marking by ensuring that the least common ancestor of two marked nodes is also marked. By definition, the graph induced by the marked leaves remains unchanged by this operation.

of what?

defined

We repeat this procedure until there is no bough of dimension greater than $N_0$ whose backbone is made of marked nodes. This process terminates because every rewrite step decreases the multiset of dimensions of boughs whose backbone is made of marked nodes.

Mael: Here it may be unclear how you define this multiset, do you consider the dimension of only maximal bough ? or every possible bough ?

To verify that $(\mathcal{T}', \mathfrak{s}', \rho)$ is well-marked, note that by construction, whenever we have marked nodes $x$ and $y$ with a non-marked node in between at level $k$, this configuration arises from a bough replacement where the required witnesses $z_1, z_2, z_3$ exist by the definition of good boughs.

Assume towards a contradiction that there exists arbitrarily long paths of non-dummy nodes in the resulting tree $\mathcal{T}'$. Remark that any path that is too long contains the backbone of a bough of dimension greater than $N_0$ by a Ramsey argument.

Mael: Maybe explain a bit more here. On what depends the length of the path, does it depends on max value of the split ?

By construction, every separating node in such a bough backbone is followed by a dummy node, so without loss of generality, one can assume that the bough backbone is made solely of marked nodes. But this contradicts the termination of the procedure. We have proven that there exists a bound on the length of paths of non-dummy nodes, concluding the proof. □

Mael: Here you would say you prove 3->2 of Lemma 4.6

We conclude this section by proving one direction of Lemma 4.5. Assume that there exists a bound on the dimension of bad boughs. By Lemma 4.6, we can transform any tree into a $L$-bounded well-marked nested tree. By Theorem 3.7, the class of such trees is well-quasi-ordered under the gap-embedding ordering. Furthermore, by Corollary 3.9, the gap-embedding ordering between marked nested trees preserves the interpretation: if $(\mathcal{T}_1, \mathfrak{s}_1, \rho_1) \leq_{\text{gap}} (\mathcal{T}_2, \mathfrak{s}_2, \rho_2)$, then the graph interpreted from $(\mathcal{T}_1, \mathfrak{s}_1, \rho_1)$ is an induced subgraph of the graph interpreted from $(\mathcal{T}_2, \mathfrak{s}_2, \rho_2)$, when both are restricted to marked leaves. Since the interpretation restricted to marked leaves coincides with the original interpretation on $\mathcal{T}_1$ and $\mathcal{T}_2$ by Lemma 4.6, this establishes the desired order-preserving property, and we conclude that the image of $\mathcal{I}$ is $\forall$-well-quasi-ordered using Fact 3.1.

## 4.2 Obstructions

In this section, we will assume that there exist bad boughs of arbitrarily large dimension, and we will leverage this to construct an infinite antichain in the image of $\mathcal{I}$. Our first remark is that there are only finitely many equivalence classes for boughs of a given level $k$ with respect to compatibility. This is the content of the following Lemma 4.7. Then, we will show that actually, one does not need arbitrarily many contexts to witness that a bough is bad. This is the content of Lemma 4.8. Combining these two lemmas, we will extract an infinite sequence of bad boughs of the same level that are pairwise compatible, and that are bad in the same context. Finally, we will leverage this to construct an infinite antichain in the image of $\mathcal{I}$, as stated in Lemma 4.9.

Mael: The proof of this lemma is sometimes unclear, but the next one that prove similar things is much more clear. You can take inspiration of the later to improve.

Lemma 4.7. *Let $k$ be a level. There are finitely many boughs of level $k$ up to compatibility.*

Proof. Let $B$ be a bough of level $k$ and $C[\square]$ be a context. We want to characterise when $C[B]$ is forward Ramseyan, and show that this characterisation only depends on finitely many parameters of $B$.

Note that since $B$ itself is forward Ramseyan, and the context $C[\square]$ is also forward Ramseyan, the only way that $C[B]$ can fail to be forward Ramseyan is one of the following:

Mael: Explain that these three possible case are because FR apply only to nodes $x \sqsubseteq y$

(1) one considers two nodes $x$ and $y$ in $C[\square]$ whose path goes across $B$ (i.e., $x$ is in the root subtree, and $y$ is in either the left or right subtree).
(2) one considers a node $x$ on the branch defined by the backbone of $B$, and $y$ in the left or right subtree of $C[\square]$.
(3) one considers a node $x$ in the root subtree of $C[\square]$, and $y$ in the bough $B$.

path (branch is from root to leaf no?)

Note that for Item 1, the forward Ramseyan condition on $B$ states that for every path crossing $B$, the monoid element of the part of the path inside $B$ must be $e$, the first idempotent element labelling $B$. Hence, this condition has finite index.

Mael: What finite index mean ?

For Item 2, the forward Ramseyan condition requires nodes to be in the same $l$-neighbourhood. But, all nodes on the branch defined by the backbone of $B$ are of split level at least $k$. Hence, for $x$ and $y$ to be in the same $l$-neighbourhood, they must be of level exactly $k$. As a consequence, this condition only depends on the (finite set) of idempotent elements labelling the bough $B$ when the context is fixed.

what is $l$

Mael: I do not understand the last sentence

Finally, for Item 3, the forward Ramseyan condition concerns a node $x$ in the root subtree of $C[\square]$, and a node $y$ in $B$, but not necessarily in the branch defined by the backbone of $B$. Here, the forward Ramseyan condition boils down to the following equations: for each node $z$ in $B$ of level $l$, let $x$ be the closest ancestor of $\square$ of level $l$ (if it exists) in $C[\square]$ and $\alpha = [\![x : \square]\!]_{\mathcal{T}}$. Furthermore, let $y$ be a descendent of $z$ in $B$ of level $l$, and $\beta$ be the $[\![z : y]\!]_{\mathcal{T}}$. Then, the forward Ramseyan condition requires that $\alpha \cdot \beta = \alpha$ and $\beta \cdot \alpha = \alpha$. Note that if this condition is satisfied, then the tree is forward Ramseyan because $C[\square]$ and $B$ are. Hence, it suffices to list all possible pairs $(\alpha, l)$ for every level $l$ appearing in $B$ to characterise this condition.

Mael: What do you mean by Ramseyan condition ?

We have proven that the forward Ramseyan condition on $C[B]$ only depends on finitely many parameters of $B$. Since $M$ is finite, there are finitely many possible values for these parameters (and for the first idempotent element labelling $B$). This concludes the proof. $\square$

Lemma 4.8. *There exists a finite set $\mathcal{F}$ of contexts $C[\square]$ such that for every bad bough $B$ of level $k$, there exists a context $C[\square] \in \mathcal{F}$ such that $B$ is a bad bough in $C[B]$.*

Proof. The proof relies on the fact that the context $C[\square]$ plays a very little role in witnessing that a bough $B$ is bad.

Let $B$ and $H$ be two compatible boughs of level $k$, and let $C[\square]$ be a context such that $\mathcal{T} = C[B]$ and $\mathcal{T}' = C[H]$ are forward Ramseyan. We want to characterise when there exists an embedding $h \colon \mathcal{I}(\mathcal{T}) \to \mathcal{I}(\mathcal{T}')$ witnessing that $B$ is a good bough in $C[B]$ with respect to $H$.

First, one only needs to consider the edges between leaves of $B$ or edges between leaves of $B$ and leaves of $C[\square]$. Indeed, the map $h$ acts as the identity on leaves of $C[\square]$, so edges are preserved there.

The fact that there are edges between leaves of $B$ only depends on their respective bough types and the value of $[\![\mathrm{root} : \square]\!]_{\mathcal{T}_{\mathrm{root}}}$. As a consequence, any two contexts sharing the same value for the path from the root to $\square$ do not change the edges between leaves of $B$ (or $H$). Since there are finitely many monoid elements, there are finitely many possible values for this path.

Let us now consider edges between leaves of $B$ and leaves of $C[\square]$. Similarly as for the leaves in boughs, one can associate to every leaf of $C[\square]$ a *context type* allowing to determine the presence of an edge. For a leaf $x$ in $C[\square]$ that belongs to the root subtree, we collect the following monoid elements: $[\![\mathrm{root} : z]\!]_{\mathcal{T}_{\mathrm{root}}}$, $[\![z : x]\!]_{\mathcal{T}_{\mathrm{root}}}$, and $[\![z : \square]\!]_{\mathcal{T}_{\mathrm{root}}}$, where $z$ is the closest ancestor of $x$ belonging to the path from root to $\square$. For a leaf that belongs to the left (resp. right) subtree, we only consider the monoid element $m_{\mathrm{left}} \cdot [\![\mathrm{root} : x]\!]_{\mathcal{T}_{\mathrm{left}}}$ (resp. $m_{\mathrm{right}} \cdot [\![\mathrm{root} : x]\!]_{\mathcal{T}_{\mathrm{right}}}$). It is immediate that to compute the presence of an edge between a leaf $y$ in $B$ and a leaf $x$ in $C[\square]$, it suffices to know the bough type of $y$ and the context type of $x$ (see Equation (1), and Figure 5). Since there are finitely many possible context types and bough types, if a bough $B$ is a good bough in $C[B]$ using a map $h$ and a compatible bough $H$, then for any other context $C'[\square]$ sharing the same context types for its leaves as $C[\square]$, the bough $B$ is also a good bough in $C'[B]$ using the same map $h$ and the same compatible bough $H$.

Since there are finitely many possible contexts up to this equivalence relation, we conclude the proof. $\square$

Mael: Say here that the following lemma prove the implication 1->3 of Lemma 4.5 and conclude its proof

Lemma 4.9. *If there exist bad boughs of arbitrarily large dimension, then there exists an infinite two-colored antichain in the image of $\mathcal{I}$.*

Proof. Assume towards a contradiction that the class of graphs obtained as images of $\mathcal{I}$ is 2-well-quasi-ordered. Let $(B_i)_{i \geq 1}$ be an infinite sequence of bad boughs. Without loss of generality, one can assume that all boughs $B_i$ have the same level $k$, and are pairwise compatible, by Lemma 4.7.

Now, using Lemma 4.8, one can extract one context $C[\square]$ such that infinitely many boughs $B_i$ are bad boughs in $C[B_i]$. Finally, one can extract further to assume that the dimension of the boughs $B_{i+1}$ is greater than twice the number of leaves in $B_i$.

Let us now color every graph $\mathcal{I}(C[B_i])$ with two labels: $\top$ to leaves belonging to $B_i$, and color $\bot$ to leaves belonging to $C[\square]$. Because we assume that the image of $\mathcal{I}$ is 2-well-quasi-ordered, we can extract an infinite increasing subsequence, and assume that for all $i < j$, there exists an embedding $f_{i,j} \colon \mathcal{I}(C[B_i]) \to \mathcal{I}(C[B_j])$ that preserves labels. Let us remark that for all $i < j$, the map $f_{i,j}$ acts as a permutation when restricted to nodes labelled $\bot$, that is of size the number of leaves in $C[\square]$. By a Ramsey argument, we can therefore assume that this permutation is the identity for all $i < j$. Finally, consider any two indices $i < j$.

Because the dimension of $B_j$ is greater than twice the number of leaves in $B_i$, there must be some bough block in $B_j$ that is left untouched by $f_{i,j}$. Furthermore, we know that $f_{i,j}$ is the identity on leaves belonging to $C[\square]$. Hence, the map $f_{i,j}$ witnesses that $B_i$ is a good bough in $C[B_i]$, which is a contradiction. $\square$

## 5 BOUNDED CLIQUE WIDTH

In this section, we will aim at leveraging the previous results to tackle classes of bounded clique-width. Our first goal is to prove our Theorem 1.5 regarding the images of finite trees under MSO-interpretations. There are two difficulties here: first, we only dealt with simple MSO-interpretations so far, and second, we have not yet given any decision procedure to check whether the image of a given simple MSO-interpretation is 2-well-quasi-ordered. Let us first tackle the second difficulty.

*Definition 5.1 (Perfect Bough).* Let $\mathcal{T} = C[B]$ be a tree with a bough $B$ of level $k$. We say that $B$ is a *perfect bough* there exists a compatible bough $H$ of level $k$ and a map $h \colon \mathcal{I}(C[B]) \to \mathcal{I}(C[H])$

(1) $h$ is an embedding of graphs,
(2) $h$ is the identity map on leaves belonging $C[\square]$
(3) the bough type of every leaf $x$ in $H$ is the same as the bough type of $h(x)$ in $B$.
(4) there exists a block in $H$ that is left untouched by $h$.

Mael: problem with last condition less restrictive than the one for good bough

An *imperfect bough* is a bough that is not a perfect bough.

Remark that every perfect bough is a good bough, hence every bad bough is an imperfect bough.

Mael: Say that you extend Lemma 4.5 to perfect bough

Lemma 5.2. *The image of $\mathcal{I}$ is 2-well-quasi-ordered if and only if the image of $\mathcal{I}$ is $\forall$-well-quasi-ordered if and only if there exists a bound on the dimension of imperfect boughs.*

Proof. First, if we have a bound on the dimension of imperfect boughs, then we also have a bound on the dimension of bad boughs, because every bad bough is an imperfect bough. Hence, by Lemma 4.5, the image of $\mathcal{I}$ is $\forall$-well-quasi-ordered.

Conversely, assume that there exist imperfect boughs of arbitrarily large dimension. Then, by an argument similar to the one of Lemma 4.9, we can construct an infinite labelled antichain in the image of $\mathcal{I}$. The only change is that one also adds the bough type of the leaves as additional labels. This actually follows the original proof technique of [17] for bounded linear clique-width. □

The important advantage of Lemma 5.2 over Lemma 4.5 is that imperfect boughs are easily recognizable using Monadic Second-Order logic on trees. Indeed, every bough $B$ is actually a (binary) tree with some distinguished nodes that form its backbone. Hence, given a tree $\mathcal{T}$ together with a split $\mathfrak{s}$ and labels on its edges coming from a finite monoid $M$, one can ask whether this tree with distinguished nodes represents an imperfect bough.

Lemma 5.3. *There exists an* MSO *formula that recognizes imperfect boughs.*

Proof Sketch. The proof follows the same pattern as [17, Lemma 19]. The main idea is that if a bough $B$ is perfect, then the only thing that matters for a given leaf $x$ in $B$ is its bough type and whether it is mapped on the left or right of the untouched bough block in the bough $H$. Hence, one can guess for every leaf in $B$ where it is mapped (left or right of the untouched block), and verify that the mapping preserves edges using only MSO formulas.

Mael: Give a bit more intuition, like the idea is to check if B and BBB are compaitble, which is MSO definable ?

t'avais juste envie de dire ce mot non?

If such a mapping exists, then one can reconstruct a compatible bough $H$ by triplicating $B$, that is, letting $H \triangleq BBB$, and mapping leaves to their corresponding copy in $H$. The middle copy will be untouched, witnessing that $B$ is a perfect bough. This reasoning was made formal for bounded linear clique-width in [17, Lemma 16], and the exact same proof works here. □

Corollary 5.4. *One can decide whether there are imperfect boughs of arbitrarily large dimension.*

Proof. We will use the theory of Cost-MSO formulas [6, Section 6]. Let us call $\Phi$ the formula of Lemma 5.3 that recognizes imperfect boughs. Let us now consider the following Cost MSO formula:

$$\Psi(N) \triangleq \Phi \wedge |\{\text{number of elements in the backbone}\}| \leq N$$

Mael: what is number of elements in the backbone, the dimension of bough, the size of the path from $b_0$ to $b_n$ ?

This formula takes an input parameter $N$ (a natural number), and decides whether the provided tree with split and labels is an imperfect bough of dimension at most $N$. By [6, Corollary 16], one can decide whether such a formula can hold for arbitrarily large values of $N$, effectively answering our question.

Mael: I am not familiar with this, but this is not obvious for me that [6, Corollary 16] actually imply this

□

We are now ready to prove our Theorem 1.5.

Mael: restate the thm?

Proof. Let $\mathcal{I}$ be an MSO-interpretation. We can get rid of the domain formula and selection formula by standard arguments without changing the 2-well-quasi-ordering status of the image of $\mathcal{I}$ [17, Lemma 8]. Hence, we can assume that $\mathcal{I}$ is a simple MSO-interpretation. Furthermore, the transformation from an simple MSO-interpretation to a monoid interpretation from trees described in Section 3 is effective. Finally, by Lemma 5.2 the image of $\mathcal{I}$ is 2-well-quasi-ordered if and only if it is $\forall$-well-quasi-ordered, and by Lemma 5.3, one can decide whether the image of $\mathcal{I}$ is 2-well-quasi-ordered.

It remains to discuss the fact that one can add a total ordering on the graphs without changing the 2-well-quasi-ordering status of the image of $\mathcal{I}$. This is because we prove that the image of $\mathcal{I}$ is 2-well-quasi-ordered using the marked gap-embedding ordering on marked nested trees representing the graphs, which already includes a total ordering on the leaves. □

Let now turn our attention to hereditary classes to obtain our Corollary 1.6.

Proof. By a standard argument, if the class $\mathcal{I}$ is hereditary and 2-well-quasi-ordered, then it is characterised by finitely many forbidden induced subgraphs [9, Proposition 3]. Hence, if $C$ has bounded clique-width and is hereditary, one can assume that $C$ is exactly the image of an MSO-interpretation $\mathcal{I}$ from finite trees, and the result follows from Theorem 1.5.

Mael: I don't see why you can assume that "$C$ is exactly the image of an MSO-interpretation"

□

Let us now turn our attention to Theorem 1.7, and in particular the link between 2-well-quasi-ordering and bounded linear clique-width. The following lemma shows that the existence of imperfect boughs of arbitrarily large dimension can be witnessed using only blocks from a finite set.

Lemma 5.5. *Assume that there are imperfect boughs of arbitrarily large dimension. Then, there exists a finite set of blocks $\mathbb{F}$ such that one can build an imperfect bough of arbitrarily large dimension using only blocks from $\mathbb{F}$.*

Proof. The proof follows the same pattern as Lemmas 4.7 and 4.8, and continues on the ideas of Lemma 5.3. The idea is that if a bough $B$ is perfect, then so is $H \triangleq BBB$. But this only depends on the bough types present in $B$. As a consequence, if $B$ and $B'$ exhibit the same bough types for their leaves, then $B$ is perfect if and only if $B'$ is perfect. If one defines an equivalence relation between bough blocks of a given bough $B$ based on the set of available bough types of its leaves, then one can always replace a bough block by another

what is this ?

one in the same equivalence class without changing the perfectness of $B$. Since this equivalence relation has finitely many classes, the result follows. □

As an immediate consequence of Lemma 5.2 and Lemma 5.5, we see that

Mael: Explain a bit more how you assemble this two lemma to obtain the corollary. I don't see where the linear clique-width comes from.

Corollary 5.6. *Assume that the class $\mathcal{I}$ is hereditary. Then, the image of $\mathcal{I}$ is 2-well-quasi-ordered if and only if every subclass of bounded linear clique-width in $\mathcal{I}$ is 2-well-quasi-ordered.*

## 6 STRUCTURAL CHARACTERISATIONS

In this section, we will be interested in extracting a very regular structure from a graph class $C$ that is not $\forall$-well-quasi-ordered. This structure will take two forms, the first one is a combinatorial notion of periodic antichain that is similar to the one proposed in [4, Section 7], and the second one is a logical property of existentially transducing the class of all finite paths.

*Definition 6.1 (Periodic Sequence).* A *periodic sequence (of graphs)* is given by a finite graph $G$, a labelling function $\ell\colon V(G) \to \Sigma$ for some finite set $\Sigma$, and two sets $C, F \subseteq \Sigma^2$. It defines an infinite set of graphs $(G^r)_{r\geq 1}$ as follows: the vertex set of $G^r$ is $V(G) \times \{1, \dots, r\}$, and there is an edge between $(u, i)$ and $(v, j)$ with $i \leq j$ if and only if one of the following holds:

- $(u, v) \in E(G)$ and $i = j$;
- $(\ell(u), \ell(v)) \in C$ and $|i - j| = 1$
- $(\ell(u), \ell(v)) \in F$ and $|i - j| > 1$.

It follows from results of the previous section that one can extract periodic antichains from any hereditary class of graphs that is not $\forall$-well-quasi-ordered.

Lemma 6.2. *If a hereditary class of graphs $C$ is not $\forall$-well-quasi-ordered, then it contains a periodic sequence $(G^r)_{r\geq 1}$ that is an antichain for the labelled induced subgraph relation.*

Proof. Assume that $C$ is not $\forall$-well-quasi-ordered, and in particular, not 2-well-quasi-ordered by Corollary 1.6. By Corollary 5.6, it contains a subclass $C'$ of bounded linear clique-width that is not 2-well-quasi-ordered. Now, following [17, 3.2 Characterizing Labelled Well-Quasi-Ordered Classes], there is an automaton that witnesses the fact that there are imperfect boughs of arbitrarily large dimension in $C'$ (they are called *bad forest paths* in [17]). By a pumping argument,

Mael: I don't know what is a pumping argument haha

one can extract a sequence of the form $G_l G^n G_r$ of imperfect boughs of arbitrarily large dimension for some fixed imperfect boughs $G_l, G_r$ and $G$. By constructions $G^n$ must be itself an imperfect bough for every $n \geq 1$. Note that we have exactly the structure of a periodic sequence here, with $C$ being the set of pairs of labels that create edges between consecutive copies of $G$, and $F$ being the set of pairs of labels that create edges between non-consecutive copies of $G$. □

Let us now leverage this combinatorial structure to extract a logical property of the class $C$.

Lemma 6.3. *Let $C$ be a periodic antichain (having the coloring defined) as in Lemma 6.2. Then $C$ existentially interprets the class of all finite paths.*

Proof.

Mael: Draft, most arguments are here, let me know if there is something that sounds fishy, i'm not used to write such things. I will rewrite it nicer later

Let $C = (G^r)_{r\geq 1}$ be a periodic antichain with labelling function $\ell\colon V(G) \to \Sigma$ and sets $C, F \subseteq \Sigma^2$ . Moreover assume that $\ell$ is injective up to duplicating labels from $\Sigma$, in particular $|\Sigma| = |V(G)|$. For $i \leq r$, we denote the $i$-th copy of $G$ in $G^r$ by $G_i$, and by $u_i \in G^r$ the $i$-th copy of $u \in V(G)$. We will proceed in two steps, we first define a formula producing a directed graph where there is a directed shortest path from $G_1$ to $G_r$ going through every $G_i$, but where there is may be arcs going from $G_i$ to $G_j$ for $j \leq i$. We will then show how to get rid of theses back arcs, and produce the path $P_r$.

We now define formula:

$$\phi_{\to}(u, v) = \neg(uv \in E(G^r) \Leftrightarrow l(u)l(v) \in F)$$

Informally, $\phi_{\to}(u, v)$ creates an arc $(u, v)$ if the edge relation between $u, v \in G^r$ differs if $u$ and $v$ are put far away form each other in this order (note that $u$ can be in a copy of $G$ that comes after the one of $v$ in $G^r$). Let $H^r = \mathcal{I}(G^r)$ for $\mathcal{I} = (\top, \phi_{\to}, \top)$, note that $H$ is a directed graph as $\phi_{\to}$ is not symmetric.

*Claim 6.4.* For $u_i v_j \in E(H^r)$ only if $j \leq i + 1$.

Proof. If $j \geq i + 1$, then $uv \in E(G^r)$ if and only if $l(u)l(v) \in F$ by definition of $G^r$. ■

*Claim 6.5.* If $u_i v_j \in E(H^r)$ for $j < i$ then for every $j' \leq i$, $u_i v_{j'} \in E(H^r)$.

Proof. Follow from the definiton. ■

A spanning path of $H^r$ is a directed shortest path from $V(H_1)$ to $V(H_r)$ going through at least one vertex from each $V(H_i)$.

*Claim 6.6.* There is a spanning path in $H^r$.

Proof. Let $V_i^+$ be the vertices of $H_i$ such that there is an arc from $H_{i-1}$ to them, $V_i^-$ be the vertices of $H_i$ such that there is an arc from them to $H_{i+1}$, and $V_i^o$ be the remaining vertices of $V_i$. Suppose there is no spanning path in $H^r$, Claim 6.4, it have to be because either there is no path in $H_i$ from $V_i^+$ to $V_i^-$, or that they are empty (i.e. there is no arc from $H_i$ to $H_{i+1}$).

Mael: Argue that in this case we can embed $G^r$ into $G^{r+1}$, contradicting the antichain.

■

Let $P$ a spanning path of $H^r$, furthermore assume that it is periodic, i.e. that $P$ go through the same sequence of vertices in each $H_i$, say $P_i = u_i^1 \dots u_i^p$, and $P = P_1 P_2 \dots P_r$. If there is a backward arc $uv \in E(H^r)$ in $P$ with $u \in P_i$ and $v \in P_j$ with $i > j$, then by

Claim 6.5 this arc exists between every copy of $u$ and $v$ where $v$ is an earlier copy than the one of $u$. Define now the formula

$$\phi_P(u,v) = \mathrm{dist}_P(u,v) \leq p \wedge \mathrm{dist}_P(v,u) \leq p \wedge l(u) = l(v) = c,$$

where $c = \ell(u_1^p)$. This formula create a path from $u_1^p$ to $u_{r-1}^p$, moreover this path is undirected as $\phi_P(u,v)$ is symmetric. Consider the interpretation $\mathcal{I} = (\top, \phi_P, \phi_{univ})$, where

$$\phi_{univ}(u) = l(u) = c \wedge \exists v, \phi_P(u,v)$$

(the exist is to remove the vertex $u_r^p$). It follows that $\mathcal{I}((G^r)_{r\geq 2})$ is the class of all path. It is easy to see that these formula are existential first order.

□

of bnd cw?

Corollary 6.7. *If a hereditary class of graphs $C$ is not $\forall$-well-quasi-ordered, then it existentially transduces the class of all finite paths.*

Proof.

**MAEL ICI**

It follows from Lemma 6.2 and Lemma 6.3. □

Mael: Proof of Corollary 6.7

## 6.1 From Transductions to Interpretations

Let us recall the definition of weak periodic antichain from [4, Conjecture 2]: given a finite alphabet $\Sigma$, and a finite set $S \subseteq \Sigma^2$, one can map a word $w \in \Sigma^*$ to a graph $G_w$ whose vertex set is the positions of $w$, and where there is an edge between two positions $i+1 < j$ if and only if $(w[i], w[j]) \in S$, and an edge between two consecutive positions $i$ and $i+1$ if and only if $(w[i], w[i+1]) \notin S$. In this setting, a *(weak) periodic sequence* is given by a finite word $u \in \Sigma^+$, and a finite set $S \subseteq \Sigma^2$, and defines the infinite set of graphs $(G_{u^r})_{r\geq 1}$. It is easy to see that every weak periodic sequence is also a periodic sequence as defined in this paper, by grouping the vertices of $u$ into a single graph $G$ and adapting colors. Finally, a *weak periodic antichain* is a weak periodic sequence that is an antichain for the labelled induced subgraph relation when the first and last letters of the words $u^r$ are given distinct colors.

We now prove that [4, Conjecture 2] holds for hereditary classes of graphs having bounded clique-width.

Lemma 6.8. *If a hereditary class of graphs $C$ is not $\forall$-well-quasi-ordered, then it contains a weak periodic antichain.*

Proof. It follows from Lemma 6.2 and the fact that every periodic sequence contains a weak periodic sequence by coloring the first and last vertices of $G$ with distinct colors. Indeed, this is what we did in the proof of Lemma 6.3. □

From this, we can now try to prove the following conjecture, which would complete the structural characterization of hereditary classes of graphs that are $\forall$-well-quasi-ordered.

Conjecture 6.9. *A hereditary class of graphs $C$ is $\forall$-well-quasi-ordered if and only if it does not existentially interprets the class of all finite paths.*

Sketch of proof. The only difficult direction is to actually interpret paths, not using colors to identify the behaviour of the nodes inside the periodic sequence.

**TODO: maybe it is possible to adapt the proof!**

□

# 7 CONCLUDING REMARKS

We have proven that for all classes of graphs of bounded clique-width, the wqo-wqo and labelled-wqo are two equivalent notions. This result is a first step towards proving this conjecture for arbitrary classes of graphs.

- A natural generalisation is to consider classes of finite relational structures that are of bounded clique-width (images of finite labelled trees under MSO interpretations). There are no main obstacles to extending our techniques, except that the *accepting parts* defining monoid interpretations become more complex to handle
- What about classes that are monadically dependent ?
- The only ordering that exists on graphs that is WQO is when they are looking like trees...

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